NEW RADIO COMMUNICATION FOR NEW ERA

NRC7394 Evaluation Kit User Guide

(CLI Application)

Ultra-low power & Long-range Wi-Fi

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NEWRACOM, Inc.

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NRC7394 Evaluation Kit User Guide (CLI Application)
Ultra-low power & Long-range Wi-Fi

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1 Overview

This document introduces NRC7394 command line interface (CLI) application. The user can utilize the CLI application to check basic information for firmware, monitor channel quality; such as: received signal strength indicator (RSSI), signal to noise ratio (SNR), adjust transmit power, and configure NRC7394 to run in specific operating condition. The source code of this application is offered to user's so users can build an executable file suitable for the host.

1.1 Software structure of CLI application

As shown in Figure 1.1, the CLI application is a user-level application program. The CLI application uses Netlink library to communicate with NRC 11ah driver running on a Linux kernel. The CLI command initiated by the user and goes to the NRC 11ah driver and then to NRC7394 via host serial peripheral interface (HSPI).

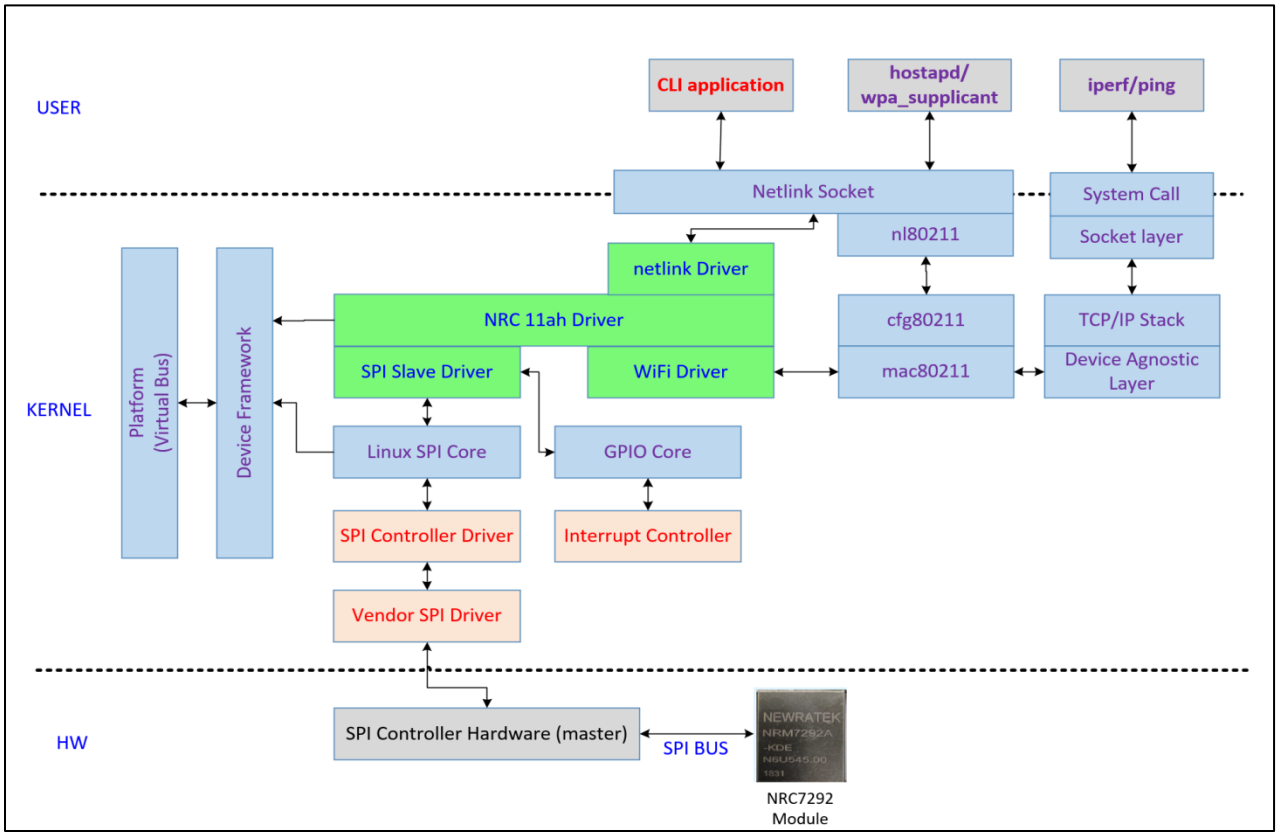


Figure 1.1 Software structure of CLI application

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1.2 Build CLI application

A user can build the source code of CLI application with Makefile which is provided along with the source code. Once the user successfully builds it, the user can find "cli_app" executable file in the same directory where the user runs Makefile.

```
pi@raspberrypi:~/cli_app $ make clean;make
libcli_app.a
cli_app
pi@raspberrypi:~/cli_app $
```

Figure 1.2 Build CLI application

1.3 Start CLI application

“NRC>” prompt appears if a user executes "cli_app", then the user can enter in a CLI command into the prompt.

```
pi@raspberrypi:~/Downloads/7292_host_codes/cli_app $ ./cli_app
================================================
    Newracom Command Line Application (2.23.0)
================================================

NRC>
```

Figure 1.3 Execution of CLI application

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“help” is a useful command which displays all the CLI commands with its usages as shown in Figure 1.4.

```
NRC> help
================================================================================================================================================
 help                                                                                  :show CLI tree
 write {address} {data}                                                                :write a 32-bit value to memory
 exit                                                                                  :exit program
 show version                                                                          :show version
 show config [vif_id]                                                                  :show configuration
 show edca                                                                             :show EDCA parameters
 show uinfo [vif_id]                                                                   :show UMAC information
 show ampdu | show ampdu clear                                                         :show/clear AMPDU count
 show signal {start|stop} [interval] [number]                                          :show rssi/snr, {options} are only valid in cli_app prompt
 show maxagg                                                                           :show max aggregation
 show duty                                                                             :show duty cycle
 show autotxgain                                                                       :show autotxgain
 show cal_use                                                                          :show cal_use
 show recovery stats                                                                   :show recovery
 show detection stats                                                                  :show detection
 show temp                                                                             :show temp
 show wakeup_pin                                                                       :show wakeup pin configuration
 show wakeup_source                                                                    :show wakeup source configuration
 show sta [vif_id] {all|aid [aid_index]}                                               :show station information
 show ap [vif_id]                                                                      :show ap information
 show tx_time                                                                          :show tx_time about {CS time} {Blank time}
 show cca_thresh                                                                       :show cca_thresh(unit: dBm)
 show self_config {Country(KR,US...)} {BW} {dwell time}                                :show self_config
 show optimal_channel {Country(US,NZ...)} {BW} {dwell time}                            :show optimal_channel
 show app_version                                                                      :show app version
 show sysconfig                                                                        :show sysconfig
 show rc [vif_id] [aid]                                                                :show tx's retry mcs info, maxtp/tp2/maxp/lowest
 show rc_pf                                                                            :show rate control profile number
 show rc_param                                                                         :show configured rate control parameter
 show xtal_status                                                                      :show xtal_status
 show stats simple_rx                                                                  :show received packet information
 show mac clear                                                                        :clear TX/RX Statistics
 show mac tx stats                                                                     :show TX Statistics
 show mac tx clear                                                                     :clear TX Statistics
 show mac rx stats                                                                     :show RX Statistics
 show mac rx clear                                                                     :clear RX Statistics
 set gi {auto|short|long} {vif_id(0|1)}                                                :set guard interval
 set maxagg {AC(0-3)} {Max(0-8(1Mhz),0-16(2,4Mhz),0:off)} {size:default=0}              :set aggregation
 set ack_mode {no|ndp|normal|show}                                                     :set ack mode
 set rc {on|off} [vif_id] [mode]                                                       :set rate control
 set duty {on|off} {duty window} {tx duration}                                         :set duty cycle
 set duty_debug {on|off}                                                               :set debug mode for duty cycle
 set txpwr {auto|limit|fixed} {value}                                                  :set tx power and type
 set wakeup_pin {Debounce(on|off)} {PIN Number(0~31)}                                  :set wakeup pin for deepsleep
 set wakeup_soruce rtc gpio hspi                                                       :set wakeup source for deepsleep
 set addba [tid] {mac address}                                                         :set addba tid / send addba with mac address
 set delba [tid] {mac address}                                                         :set delba tid / send delba with mac address
 set rts {on|off|default} <threshold> <vif_id> {ndp rts ri:1, normal rts ri:2}          :set rts on/off
 set cts {adaptive cts:on, legacy cts:off}                                             :set cts on/off
 set tx_time {CS time} {Blank time}                                                    :set tx_time about {CS time} {Blank time}
 set drop [vif id] [mac address] {on|off}                                              :set drop frames from configured mac address
 set self_config {Country(KR,US...)} {BW} {dwell time}                                 :set self_config
 set tsensor [GPIO for SCL] [GPIO for SDA]                                             :set temperature sensor scl, sda
 set cca_thresh {CCA threshold(unit:dBm, -100~-35)}                                    :set cca threshold
 set color {value}                                                                     :set color
 set deepsleep_gpio {dir} {out} {pullup}                                               :set GPIO direction/out data/pull during deepsleep operatio
n
 set report {on/off}                                                                   :set lmac periodic report
 set support_ch_width [0|1]                                                            :set supported ch width in s1g capa ie (0:1/2M, 1:1/2/4M)
 set ampdu_mode [disable|manual|auto]                                                  :set ampdu_mode
 set bcn_mcs [vif_id] [10|0|1|2|3|4|5|6|7]
                                  :set bcn_mcs
 set rc_pf [1|2]                                                                       :set rate control profile number
 set rc_param {1|2|3|4|5} {1|2|3|4|5|6|7}                                              :set rate control parameter
 set bgscan_trx [1:enable|0:disable] [wait time operation ch for rx: (0~100)msec]:set bgscan_trx
 set scan_period [dwell time (min 20ms)]                                               :set scan_period
 set mesh_rssi_threshold {-120~-10dBm}                                                 :set mesh_rssi_threshold
 test mcs [mcs index]                                                                  :test mcs
 test country [country code]                                                           :test country
 test cont_tx {stop} | {freq(in MHz)} {bw(1m|2m|4m)} {mcs} {txpwr}                     :test continuous tx
 gpio read [pin number]                                                                :gpio read
 gpio write [pin number] [0|1]                                                         :gpio write
 gpio direction [pin index] {[0(input)|1(output)]}                                     :read/write gpio direction
 gpio pullup [pin index] {[0(off)|1(on)]}                                              :read/write gpio pullup enable|disable
================================================================================================================================================
OK
```

Figure 1.4 “help” CLI command

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To stop the CLI application, a user can use “exit” command.

```
NRC> exit
================================================
      Exit Newracom Command Line Application
================================================
pi@raspberrypi:~/nrc_pkg/script $
```

Figure 1.5 Exit of CLI application

2 CLI Commands

The categories of CLI commands is described in the below. The user can use the phy, set, and test CLI commands only for test purposes.

Table 2.1 Category of CLI commands

Category	Description
show	display statistics, status, signal, etc.
set	set MAC-layer parameters
test	set test parameters
write	Write a 32-bit value to memory
gpio	write/read gpio, set gpio dirction & pullup

2.1 Show Commands

2.1.1 show version

Display Firmware version and Gerrit/master number.

Parameters

N/A

```
NRC> show version
Newracom Firmware Version        : 01.04.00
gerrit/master                    : e1d9236
OK
```

2.1.2 show uinfo [vif_id]

Display 11ah capability information.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

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```
NRC> show uinfo 0
-----------------------------------------------------|* AP INFO *|-----------------------------------------------------
[    0]  bssid(00:00:00:00:00:00)       ssid()  ssid_len(0)     security(0)     beacon_interval(0)
         short bi(0)                    assoc_s1g_channel(0)    cssid(0x0)      change_seq_num(0)
support: s1g_long(0)    pv1(0)          nontim(0)               twt(0)          ampdu(0)
         ndp_pspoll(0)                  traveling pilot(0)      shortgi(1mhz:0, 2mhz:0, 4mhz:0) 1m_ctrl_resp_preamble(0)
         maximum mpdu_len(0)            ampdu_len_exp(0)        minimum mpdu_start_spacing(0)   rx_s1gmcs_map(0x0)      color(0)

-----------------------------------------------------|* STA INFO *|-----------------------------------------------------
[    0]  mac_addr(8c:0f:fa:00:29:01)    aid(0)                  listen_interval(0)
support: s1g_long(1)    pv1(0)          nontim(1)               twt(0)                          ampdu(1)
         ndp_pspoll(0)                  traveling_pilot(0)      shortgi(1mhz:0, 2mhz:0, 4mhz:0) 1m_ctrl_resp_preamble(0)
         maximum mpdu_len(0)            ampdu_len_exp(1)        minimum mpdu_start_spacing(0)   rx_s1gmcs_map(0xfd)

-------------------------------------------------------------------------------------------------------------------------
OK
```

2.1.3 show config [vif_id]

Display device configurations including device mode, MAC address, frequency, bandwidth, etc.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

```
NRC> show config
-----------------------------------------------------
[MAC Configuration]
Boot Mode                       : ROM_CM3
Device Mode                     : STA
MAC Address                     : 8c:0f:bb:0a:cc:01
Country                         : US
Bandwidth                       : 1M
 - PRI CH BW                    : 1M
 - PRI CH LO                    : 1
 - PRI CH NUM                   : 37
 - Center LO                    : ON
Frequency                       : 9205 (37)
MAC80211_freq                   : 5240 (48)
Default MCS                     : 2
Rate Control                    : ON
 - Mode                         : Feedback RC
 - Info                         : MCS:7 BW:1 Mhz (NRC Auto)
 - MCS10(MGMT)                  : ON
Guard Interval                  : AUTO
Security                        : OFF
RTS                             : OFF
CTS                             : RID
Format                          : S1G
Preamble type                   : S1G_1M
Promiscuous Mode                : OFF
Color                           : 0x0
Auto CFO Cal                    : ON
TX Retry Limit                  : 8
Fragment                        : ON
Defragment                      : ON
PS pretend                      : OFF
Power Save                      : OFF
BSSID                           : 44:22:33:11:22:33
AID                             : 1
Scan Type                       : NORMAL
Scan Mode                       : ACTIVE

[PHY Configuration]
Tx Gain                         : AUTO
Base Rx Gain                    : 65
Tx Power Type                   : ~24 [LIMIT]
-----------------------------------------------------
OK
```

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2.1.4 show ampdu [clear]

Display statistics for aggregated MPDU (A-MPDU).

Parameters

clear : clear all statistics

```
NRC> show ampdu
-------------------
  AMPDU : Value
-------------------
 1      : 831
 2      : 0
 3      : 0
 4      : 0
 5      : 0
 6      : 0
 7      : 0
 8      : 0
 9      : 0
10      : 0
11      : 0
12      : 0
13      : 0
14      : 0
15      : 0
16      : 0
-------------------
OK
```

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2.1.5 show edca

Display the Enhanced Distributed Channel Access (EDCA) parameters per access category (AC).

Parameters

N/A

```
NRC> show edca
-------------------------------------------------
[AC]                            : 0
 - priority                     : 0
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 2
 - cw min                       : 16
 - cw max                       : 1024
 - txop limit                   : 0
 - txop max                     : 0

[AC]                            : 1
 - priority                     : 1
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 2
 - cw min                       : 16
 - cw max                       : 1024
 - txop limit                   : 0
 - txop max                     : 0

[AC]                            : 2
 - priority                     : 2
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 2
 - cw min                       : 16
 - cw max                       : 1024
 - txop limit                   : 0
 - txop max                     : 0

[AC]                            : 3
 - priority                     : 3
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 2
 - cw min                       : 16
 - cw max                       : 1024
 - txop limit                   : 0
 - txop max                     : 0

[AC]                            : 4
 - priority                     : 4
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 1
 - cw min                       : 1
 - cw max                       : 1
 - txop limit                   : 0
 - txop max                     : 0

[AC]                            : 5
 - priority                     : 5
 - aggregation                  : 0
 - max agg num                  : 16
 - aifsn                        : 3
 - cw min                       : 16
 - cw max                       : 1024
 - txop limit                   : 0
 - txop max                     : 0

-------------------------------------------------
OK
```

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2.1.6 show sta [vif id] <all|aid <aid index>>

Display station information including TX/RX PHY rate.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

all | aid <aid index> : 'all' shows information of whole connected stations. 'aid <aid index>' shows information of the specified station.

```
NRC> show sta 0 all
STATION    MAC_ADDR            AID    STATE     TX bit rate        RX bit rate
================================================================================
0          ca:fe:ba:be:01:33   1      ASSOC   6.50MBit/s(MCS 7)  6.50MBit/s(MCS 7)
Duplicate AID count : 0
================================================================================
OK
```

2.1.7 show ap [vif id]

Display ap information including TX/RX PHY rate.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

```
NRC> show ap 0
        BSSID          BW          TX bit rate        RX bit rate
=====================================================================
8c:0f:fa:00:29:01      2M        5.20MBit/s(MCS 5)   6.50MBit/s(MCS 7)
=====================================================================
OK
```

2.1.8 show signal [start|stop] [interval] [number]

Display channel quality information (RSSI and SNR).

Parameters

start : start periodic display with interval (Ex. show signal start 1: display RSSI & SNR every 1 second)
stop : stop displaying
interval : period in second unit (default 1 second)
number : number of samples to display

Returns

```
NRC> show signal
MAC addr : 8c:0f:fa:00:2b:0e    rssi     : 9            snr      : 25
OK
```

※ 'show siganl' and 'show stop' could not support in one line command operation.

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```
NRC> show signal start
OK
NRC> Mac Addr : 8c:0f:fa:00:2b:0e       rssi: 9         snr: 26

Mac Addr : 8c:0f:fa:00:2b:0e    rssi: 9         snr: 26

Mac Addr : 8c:0f:fa:00:2b:0e    rssi: 9         snr: 26

Mac Addr : 8c:0f:fa:00:2b:0e    rssi: 9         snr: 26

show signal stop
-------------------------------------------------
[MAC Addr]: 8c:0f:fa:00:2b:0e
[Total]   : 23
[RSSI]
 average  : 1.565
 std_dev  : 3.411
[SNR]
 average  : 4.522
 std_dev  : 9.855
-------------------------------------------------
OK
```

Total : total number of samples displayed,
average : average value of RSSI and SNR
std_dev : standard deviation

2.1.9 show maxagg

Display aggregation status per AC.

Parameters

N/A

```
NRC> show maxagg
------- VIF0 -------
[Base info]
AC_BK: OFF (16,    0 bytes)
AC_BE: OFF (16,    0 bytes)
AC_VI: OFF (16,    0 bytes)
AC_VO: OFF (16,    0 bytes)
OK
```

2.1.10 show cca_thresh

Display CCA threshold value.

Parameters

N/A

```
NRC> show cca_thresh
-70
OK
```

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2.1.11 show duty

Show status of duty cycle function.

If it is on, it shows duty window, tx duration, remain tx duration time in usec and duty error count.

Parameters

N/A

Returns

```
NRC> show duty
Duty cycle                    : off
OK
```

or

```
NRC> show duty
Duty cycle                    : on
Duty window                   : 60000000
Tx duration                   : 5000000
Remain tx duration            : 3375200
Duty error                    : 0
OK
```

2.1.12 show autotxgain

Show status of autotxgain function.

If it is on, it shows Tx power index for each MCS (Modulation Coding Scheme).

Parameters

N/A

```
NRC> show autotxgain
Auto txgain                   : on
Tx power index for MCS 0      : 23
Tx power index for MCS 1      : 22
Tx power index for MCS 2      : 22
Tx power index for MCS 3      : 19
Tx power index for MCS 4      : 19
Tx power index for MCS 5      : 17
Tx power index for MCS 6      : 17
Tx power index for MCS 7      : 16
Tx power index for MCS 10     : 23
OK
```

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2.1.13 show recovery stats

The count statistics of recovery function entered.

Parameters

N/A

```
NRC> show recovery stats
-------------------------------------------------
Number of Recovery Count : 0
Number of RX Frame regarding RX Buffer discard : 0
SN missing by QM
-------------------------------------------------
QM[#]    Missing Count          Max diff
-------------------------------------------------
QM[0]          0                 0
QM[1]          0                 0
QM[2]          0                 0
QM[3]         30                17
QM[4]          1                48
QM[5]          0                 0
-------------------------------------------------
OK
```

2.1.14 show detection stats

The count statistics of detection function entered, which are tx triggered.

Parameters

N/A

```
NRC> show detection stats
-------------------------------------------------
Tx Triggered Detection Count : 0
Rx Triggered Detection Count : 0
-------------------------------------------------
 --Non-zero Length NDP:  0
 --IP Length:            0(0)
 --MPDU Length Mismatch: 0
 --Buffer Mismatch:      0
 --MPDU Length Size:     0
-------------------------------------------------
OK
```

2.1.15 show temp

The temperature of temperature sensor. If temperature sensor is not existed, it displays Not Support'.

Parameters

N/A

```
NRC> show temp
Temperature                    : 0x1A(26)
OK
```

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2.1.16 show tx_time

Show tx_time parameters.

Parameters

N/A

```
NRC> show tx_time
CS time                         : 988
TX delay                        : 100
OK
```

2.1.17 show wakeup_pin

Get configuration of wakeup gpio pin from deep sleep mode.

Parameters

N/A

```
NRC> show wakeup_pin
Debounce : off                  Pin number : 11
OK
```

2.1.18 show wakeup_source

Get configuration of wakeup source from deep sleep mode.

Parameters

N/A

```
NRC> show wakeup_source
Wakeup source                   : RTC GPIO HSPI
OK
```

2.1.19 show xtal_status

Show indication whether the crystal oscillator is stable or not.

Parameters

N/A

Returns

```
NRC> show xtal_status
XTAL status                     : 1
OK
```

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2.1.20 show app_version

Show cli appication version.

Parameters

N/A

```
NRC> show app_version
2.19.0
OK
```

2.1.21 show stats simple_rx

Display received packet information.

Parameters

N/A

Returns

RSSI : received signal strength indication

CS_Cnt : number of carrier sense counted

PSDU_Succ : number of PSDU count successfully received (SIG CRC OK)

MPDU_Rcv : number of MPDU count received

MPUD_Succ : number of MPDU count successfully received (FCS OK)

SNR : signal to noise ratio

※ The PSDU_Succ counts NDP packets. However, the MPDU_Rcv does not count NDP packet. In addition, The PSDU_Succ regards A-MPDU packet as 1 packet.

```
NRC> show stats simple_rx
-------------------------------------------------
RSSI                            : -27
CS_Cnt                          : 595182
PSDU_Succ                       : 2603
MPDU_Rcv                        : 1134
MPDU_Succ                       : 1046
SNR                             : 31
-------------------------------------------------
OK
```

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2.1.22 show mac clear

Clear MAC-layer TX and RX statistics.

Parameters

N/A

```
NRC> show mac clear
success
OK
```

2.1.23 show mac tx stats

Display MAC-layer TX statistics.

Parameters

N/A

Returns

```
NRC> show mac tx stats
--------------------------------------------------------------------------
 MAC TX Statistics (OK count:24728, RTX count:259, last MCS:0)
--------------------------------------------------------------------------
- AC[BK]        : OK(          0/          0)  RTX(          0/          0)
- AC[BE]        : OK(        228/      28426)  RTX(        214/      25084)
- AC[VI]        : OK(          0/          0)  RTX(          0/          0)
- AC[VO]        : OK(        823/      84826)  RTX(         45/       4565)
- AC[BC]        : OK(      23677/     805039)  RTX(          0/          0)
- AC[GP]        : OK(          0/          0)  RTX(          0/          0)
--------------------------------------------------------------------------
- TYPE[MGMT]    : OK(        820/      83730)  RTX(         45/       4565)
- TYPE[CTRL]    : OK(          0/          0)  RTX(          0/          0)
- TYPE[DATA]    : OK(        231/      29522)  RTX(        214/      25084)
- TYPE[BEAC]    : OK(      23678/     805066)  RTX(          0/          0)
--------------------------------------------------------------------------
- MCS[ 0]       : OK(        168/      19824)  RTX(         36/       4248)
- MCS[ 1]       : OK(          0/          0)  RTX(          0/          0)
- MCS[ 2]       : OK(         37/       6168)  RTX(          0/          0)
- MCS[ 3]       : OK(          0/          0)  RTX(          0/          0)
- MCS[ 4]       : OK(          0/          0)  RTX(         59/       6962)
- MCS[ 5]       : OK(          0/          0)  RTX(         55/       6490)
- MCS[ 6]       : OK(          0/          0)  RTX(          4/        472)
- MCS[ 7]       : OK(          0/          0)  RTX(         57/       6726)
- MCS[10]       : OK(      24525/     892353)  RTX(         48/       4751)
--------------------------------------------------------------------------
OK
```

OK (number of packets successfully transmitted / total aggregated bytes successfully transmitted)

RTX (number of packets retransmitted / total aggregated bytes retransmitted)

※ AC[GP] : GP stands for general purpose. This is used to send a frame which is the highest priority. This is a vendor-specific function.

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2.1.24 show mac rx stats

Display MAC-layer RX statistics.

Parameters

N/A

Returns

```
NRC> show mac rx stats
-------------------------------------------------------------------
 MAC RX Statistics (OK count:4673, NOK count:12, last MCS:10)
-------------------------------------------------------------------
- AC[BK]        : OK(         0/         0) NOK(         0/         0)
- AC[BE]        : OK(      3778/   5319311) NOK(        12/     16672)
- AC[VI]        : OK(         0/         0) NOK(         0/         0)
- AC[VO]        : OK(       895/     38776) NOK(         0/         0)
- AC[BC]        : OK(         0/         0) NOK(         0/         0)
- AC[GP]        : OK(         0/         0) NOK(         0/         0)
-----------------------------------------------------------------------
- TYPE[MGMT]    : OK(       889/     38332) NOK(         0/         0)
- TYPE[CTRL]    : OK(         0/         0) NOK(         0/         0)
- TYPE[DATA]    : OK(      3784/   5319755) NOK(        12/     16672)
- TYPE[BEAC]    : OK(         0/         0) NOK(         0/         0)
-----------------------------------------------------------------------
- MCS[ 0]       : OK(        58/     17980) NOK(         0/         0)
- MCS[ 1]       : OK(         1/       130) NOK(         0/         0)
- MCS[ 2]       : OK(        21/     31710) NOK(         0/         0)
- MCS[ 3]       : OK(         5/      6114) NOK(         0/         0)
- MCS[ 4]       : OK(        29/     43790) NOK(         0/         0)
- MCS[ 5]       : OK(        11/     12318) NOK(         1/      1510)
- MCS[ 6]       : OK(       137/    206870) NOK(         3/      4530)
- MCS[ 7]       : OK(      3488/   4997538) NOK(         7/     10570)
- MCS[10]       : OK(       923/     41637) NOK(         1/        62)
-----------------------------------------------------------------------
OK
```

OK (number of packets successfully received / total aggregated bytes successfully received)

NOK (number of packets received but discarded / total aggregated bytes discarded)

※ NOK : Even though the MPDU is successfully received, it can be discarded due to duplication or address mismatch.

2.1.25 show mac tx clear

Clear MAC-layer TX statistics.

Parameters

N/A

```
NRC> show mac tx clear
success
OK
```

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2.1.26 show mac rx clear

Clear MAC-layer RX statistics.

Parameters

N/A

```
NRC> show mac rx clear
success
OK
```

2.1.27 show self_config <Country> <BW> <dwell time>

Show self config.

Parameters

Country : Country Code. KR, US, EU……

BW: scan channel bandwidth (1m, 2m, 4m)

dwell time: scan time for CCA (10 ~ 1000 ms)

```
NRC> show self_config US 1m 100
        Frequency       CCA     bandwidth
--      902.5 MHz       21.7%   1M
--      903.5 MHz       0.0%    1M
--      904.5 MHz       0.0%    1M
--      905.5 MHz       0.0%    1M
--      906.5 MHz       0.0%    1M
--      907.5 MHz       0.0%    1M
--      908.5 MHz       1.8%    1M
--      909.5 MHz       2.1%    1M
--      910.5 MHz       0.0%    1M
--      911.5 MHz       0.0%    1M
--      912.5 MHz       0.0%    1M
--      913.5 MHz       0.0%    1M
--      914.5 MHz       4.6%    1M
--      915.5 MHz       2.7%    1M
--      916.5 MHz       0.9%    1M
--      917.5 MHz       0.0%    1M
--      918.5 MHz       3.1%    1M
--      919.5 MHz       3.6%    1M
--      920.5 MHz       23.3%   1M
--      921.5 MHz       27.9%   1M
--      922.5 MHz       4.2%    1M
--      923.5 MHz       3.7%    1M
--      924.5 MHz       10.3%   1M
--      925.5 MHz       10.3%   1M
--      926.5 MHz       0.8%    1M
--      927.5 MHz       6.3%    1M
[Optimal freq.] 903.5 MHz (CCA:0.0%, BW:1M)
[*]ch_num:3
OK
```

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2.1.28 show chmgr

Show the channel manager status. The channel manager runs on an AP.

At boot it measures every channel the country allows and starts the BSS on the clearest one. While the AP runs it keeps measuring the candidates and moves the BSS to a cleaner channel once one stays better for long enough.

The first line is the boot selection: state, mode, number of candidates, bandwidth in MHz, selected channel number, S1G centre frequency in units of 0.1 MHz, busy ratio of that channel in percent and the watchdog reset count.

The second line is the runtime state: channel switching state, current S1G frequency, how many times in a row a candidate has beaten the current channel against how many are required, and the number of evaluations, channel switches and cancelled switches.

The third line lists every candidate as frequency(busy ratio), best first. The channel in use is marked with an asterisk.

Parameters

N/A

Returns

```
NRC> show chmgr
state=running mode=auto ncand=25 bw=2 pick=2 s1g=9030 cca=4.8 wdt=0
dcs=1(monitor) cur=9030 streak=0/10 eval=14 sw=0 veto=0 req=0 last=9040
*903.0(1.8)907.0(4.2)917.0(4.7)906.0(5.2)927.0(5.7)908.0(5.7)914.0(6.0)904.0(6.8)905.0(7.1)918.0(7.1)
926.0(7.3)915.0(7.4)924.0(7.7)921.0(8.0)916.0(9.5)909.0(10.2)920.0(12.4)922.0(13.9)913.0(16.0)
919.0(16.4)923.0(19.9)910.0(21.0)912.0(26.8)925.0(28.1)911.0(29.5)
OK
```

2.1.29 show cal_use

Show whether calibration data is used and country code.

Parameters

N/A

Returns

```
NRC> show cal_use
Calibration_use : on                    Country  : US
OK
```

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2.1.30 show sysconfig

Show NRC WiFi board system configurations.

Parameters

N/A

Returns

```
NRC> show sysconfig
[sysconfig]
 version        : 2
 mac_addr0      : 8c:0f:fa:00:2b:9a
 mac_addr1      : 8c:0f:fa:00:52:aa
 cal_use        : 1
 hw_version     : 65535
 rf_pllldo12_tr : 0xA5A5A5A5 (Disabled)


[user_factory]


OK
```

2.1.31 show bcn_mcs [vif id]

Show the MCS (Modulation and Coding Scheme) level for beacon transmissions.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

Returns

```
NRC> show bcn_mcs 0
------------------------ current beacon rate --------------------------
VIF#:0, Current beacon mcs = 10

-----------------------------------------------------------------------
OK


NRC> show bcn_mcs 1
------------------------ current beacon rate --------------------------
VIF#:1, Current beacon mcs = 0

-----------------------------------------------------------------------
OK
```

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2.1.32 show rc_param

Show rate control parameters.

EWMA : The percentage of Exponentially Weighted Moving Average (EWMA) used in rate control

Update interval : shows the frequency at which EWMA data is updated (unit : ms)

Probe interval: shows the probing interval (unit : ms)

Parameters

N/A

Returns

```
NRC> show rc_param
EWMA(%)                         : 30
Update interval(ms)             : 500
Probe interval(ms)              : 50

OK
```

2.1.33 show rc [vif_id] [aid]

Show rate control key values for TX retry order

maxtp : 1st TX throughput rate

tp2 : 2nd TX throughput rate

maxp : 1st TX success probability rate

lowest : Lowest throughput rate

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

aid : Association ID. AP mode needs STA's aid#. Please use 'show uinfo [vif_id]' to check the STA's aid. This can be omitted in STA mode.

Returns

[STA mode]

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```
NRC> show rc
----------------------------------------
 order                              mcs bw
----------------------------------------
 maxtp                             :  7  1
 tp2                               :  7  1
 maxp                              :  6  1
 lowest                            : 10  1

----------------------------------------
OK
```

[AP mode]

```
NRC> show rc 0 1
----------------------------------------
 order                              mcs bw
----------------------------------------
 maxtp                             :  7  2
 tp2                               :  7  2
 maxp                              :  6  2
 lowest                            : 10  1

----------------------------------------
OK
```

2.1.34 show clock_count {xtal|lpo}

Displays the clock count (Hz) for 1 second

Parameters

xtal : XTAL (crystal oscillator) clock count

lpo : LPO (low-power oscillator) clock count

Returns

```
NRC> show clock_count xtal
count                              : 32766
OK


NRC> show clock_count lpo
count                              : 29997
OK
```

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2.1.35 show rx_count

Display the total number of received (RX) packets

Parameters

None

Returns

RX packet count

```
NRC> show rx_count
13
OK
```

2.1.36 show rx_clear

Clear all RX packet statistics.

Parameters

None

Returns

RX packet count

```
NRC> show rx_clear
0
OK
```

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2.2 Set Commands

2.2.1 set gi <short|long> [vif_id]

Set Guard Interval.

Parameters

short : Short Guard Interval

long : Long Guard Interval (default)

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

```
NRC> set gi short
OK
```

2.2.2 set maxagg <ac> <maxagg> {options}

Set maximum number of MPDU of AC in A-MDPU.

(AP)

set maxagg ac maxagg // set maxagg for all-sta

set maxagg ac maxagg maxsize // set maxagg and maxsize for all-sta

set maxagg ac maxagg vif aid // set maxagg for specifi aid

(STA)

set maxagg ac maxagg //set maxagg

set maxagg ac maxagg maxsize //set maxagg and maxsize

Parameters

ac : access category

maxagg : number of MPDU in AMPDU (0-8(1Mhz), 0-16(2,4Mhz), 0: off)

maxsize: aggregation threshold size (default : 0)

vif : virtue interface identification

aid: association identification

```
NRC> set maxagg 0 2
-- updated aggregation
[STA AID:    0]
AC(BK): STATUS(ON) MAXNUM( 2) SIZE(0)
OK
```

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2.2.3 set ack_mode {mode}

Set ACK mode.

Parameters

mode: no|ndp|normal|show

```
NRC> set ack_mode no
ACK_MODE                        : NO

OK
```

2.2.4 set rc <on|off> [vif_id]

Set rate control on / off.

Parameters

on : enable rate control (automatic selection of MCS based on the link condition) (default)

off : disable rate control (user can select MCS manually by using “test mcs” command)

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

```
NRC> set rc on
------------------------ updated rate control -------------------------
rc       : on           vif_id   : 0            mode     : NRC
-----------------------------------------------------------------------
OK
```

2.2.5 set duty <on|off> {duty window in usec} {tx duration in usec}

Set duty cycle.

Parameters

on|off : enable|disable duty cycle function (default; off)

duty window : specify duty cycle window in usec. (should be >= 1000000)

tx duration : specify allowed tx duration within duty cycle window in usec. (should be >=100000)

(ex) duty window 10000000 (10sec) and tx duration 1000000 (1sec) will be set to access the channel 1 second during every 10 seconds.

```
NRC> set duty on 10000000 1000000
Duty cycle                      : on
Duty window                     : 10000000
Tx duration                     : 1000000
OK
```

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※ The supported duty cycle ratio ((Tx duration/Duty window) * 100) should be smaller than 10%. (See ETSI EN 304 220-1) But if type is STA and country code is EU, ratio should be smaller than 2.8%. (See ETSI EN 304 220-2)

2.2.6 set duty_debug <on|off>

Set duty cycle debug on/off.

Parameters

on|off : enable|disable duty cycle debug

```
NRC> set duty_debug on
duty debug : on
OK
```

2.2.7 set cca_thresh <value>

Set CCA threshold value.

Parameters

value: CCA threshold value (-100 dbm ~ -35 dbm)

```
NRC> set cca_thresh -70
-70
OK
```

2.2.8 set txpwr <auto|limit|fixed> <value>

Set tx power type and value.

Parameters

type :

auto: The device will automatically adjust its Tx power based on the current network conditions and signal strength.

limit : The device will use a specified maximum Tx power limit.

fixed : The device will use a fixed Tx power level, which can be useful for testing or for applications where a consistent power level is required.

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value : 1~30

```
NRC> set txpwr fixed 17
Type     : fixed        Tx power : 17
                        OK
```

2.2.9 set tpc <on [-f]|off|show|stats [clear|on|off]>

Set transmit power control (TPC) and show its state and statistics.

With TPC on, the device lowers its transmit power while the link has margin to spare, judged from the SNR the peer reports back. It reduces interference and power consumption in strong signal conditions, and raises the power again as soon as the link degrades.

Both ends have to support TPC. A STA learns whether the connected AP supports it during association, so Peer TPC is only meaningful once the connection is complete. TPC is not supported in MESH mode.

Parameters

on : enable TPC. -f forces it on without the capability exchange, for a STA connected to an AP that does not advertise TPC support.

off : disable TPC.

show : show the role of this device and its own TPC state. On a STA the TPC support of the connected AP is shown as Peer TPC. An AP serves many STAs, so it does not report a peer.

stats : show the TPC statistics. clear resets the counters, and on|off turns the collection on or off.

Returns

```
NRC> set tpc on
OK

NRC> set tpc show
Role                            : AP
TPC                             : on
OK
```

On a STA the TPC support of the connected AP is shown as well.

```
NRC> set tpc show
Role                            : STA
TPC                             : on
Peer TPC                        : on
OK
```

set tpc stats shows the transmit power the device settled on and how much power it saved.

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```
NRC> set tpc stats
tpc: 0 STATS ---
 state_enter NONE:0 INIT:2 WAIT:1 READY:176 SET:175
 init_reason initial:1 no_ack:0 hi_retry:0 wait_noack:1
 txdone 179  no_ack 0  ack_rate 100.0%
 retry  0:175 1:2 2:2 3:0 4+:0
 snr    samples:1 last:29 avg:29 min:29 max:29
 txp    samples:178 last:12 avg:13 min:10 max:25
 txp_hist 10:37 11:10 12:20 13:22 14:20 15:20 16:20 17:10 18:10 19:8 25:1
 floor_hit:37 (20.7%)  ceil_hit:1 (0.5%)
 saving avg:6.0 dB  total:1069 dBm-pkt
 apply  total:175  cand:1 cand_up:0 retry_up:2 stable_down:16 stable_keep:156, low_mcs:0 rst_noack:0 rst_hretry:0
OK
```

2.2.10 set wakeup_pin <debounce> <pin index>

Configure a wakeup-gpio-pin when system state is uCode or deepsleep.

Parameters

debounce : on|off

pin index : 0~31

```
NRC> set wakeup_pin off 18
Debounce : off                  Pin number : 18
OK
```

2.2.11 set wakeup_source <wakeup_sources>

Configure wakeup sources when system state is deepsleep.

Parameters

wakeup_sources : rtc gpio hspi

※ It is possible to assign multiple sources (ex) set wakeup_source rtc gpio

```
NRC> set wakeup_source rtc gpio
Wakeup source                   : rtc gpio
OK
```

2.2.12 set addba [tid] {mac address}

Set addba tid / send addba with MAC address.

Parameters

tid: Traffic Identifier

mac address : mac address

```
NRC> set addba 1
OK
```

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2.2.13 set delba [tid] {mac address}

Set delba tid / send delba with MAC address.

Parameters

tid: Traffic Identifier

mac address : mac address

```
NRC> set delba 1
OK
```

2.2.14 set rts <on|off|default> <threshold> <vif_id>

Enable/disable RTS, set RTS threshold.

Parameters

threshold : RTS & CTS threshold in Byte

vif_id : virtue interface identification

Ex1) send RTS regardless of packet length (set rts on 0 0)

Ex2) no use of RTS (set rts off 0 0)

Ex3) set RTS threshold (set rts default 1000 0)

```
NRC> set rts on 0 0
OK
```

2.2.15 set tx_time <CS time> <Blank time>

Set tx_time about <CS time> <Blank time>. CS time at least [(AFISN offset * 16 + 1) * 52us]

Parameters

CS time: Carrier sensing time. Listen before talk (52~13260 in us)

Blank time: Tx pause time (1~65535 in us)

```
NRC> set tx_time 52 10000
OK
```

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2.2.16 set drop [vif id] [mac address] {on|off}

Set drop frames from configured MAC address.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

mac address: drop frames from mac address

on|off: Enable disable drop mac address

2.2.17 set tsensor [GPIO for SCL] [GPIO for SDA]

Set temperature sensor scl, sda.

Parameters

SCL: GPIO for SCL (default 31)

SDA: GPIO for SDL (default 30)

```
NRC> set tsensor 31 30

OK
```

2.2.18 set chmgr <off|auto|monitor> [profile=0|1|2] [cc=CC] [bw=1|2|4] [exclude=<s1g,..>] | set chmgr log <0|1>

Set the channel manager mode and run the channel selection again with the given options. The channel manager runs on an AP.

off : stop the channel manager. The AP keeps the channel it is on.

auto : measure the candidates, start on the clearest channel and keep switching to a better one while the AP runs.

monitor : measure the candidates and keep scoring them, but never switch the channel.

Parameters

profile : how much better a candidate has to be before the AP moves. 0 (conservative), 1 (normal) or 2 (aggressive). The current profile is kept when this is omitted.

cc : country code the candidate list is built from (US, JP, K1 ...). The current country is kept when this is omitted.

bw : candidate channel bandwidth in MHz. 1, 2 or 4.

dwell : measuring time per channel in msec for the boot scan. 100 ~ 1000, default 300.

exclude : comma-separated S1G frequencies to leave out, in units of 0.1 MHz.

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log : 1 turns the channel manager debug log on, 0 turns it off.

(ex) set chmgr auto cc=US bw=2 exclude=9025,9035 excludes 902.5 MHz and 903.5 MHz from the candidates.

Returns

chmgr: ok pick=<channel number> when the selection succeeded, chmgr: ok off when the channel manager was stopped, and chmgr: fail(<code>) when no usable candidate was found.

```
NRC> set chmgr auto cc=US bw=2 dwell=300
chmgr: ok pick=2
OK

NRC> set chmgr monitor
chmgr: ok pick=2
OK

NRC> set chmgr off
chmgr: ok off
OK

NRC> set chmgr log 1
chmgr: log=1
OK
```

2.2.19 set color {value}

Set color bit.

Parameters

value : 0 -7

```
NRC> set color 1
OK
```

※ We just added a value to the beacon frame in our Wi-Fi setup, but it is currently not being utilized.

2.2.20 set report <on|off>

Set LMAC periodic report on/off.

Parameters

on|off : enable|disable

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```
NRC> set report on
set report : on
OK
```

[Target conole log after 'set report on']

```
-------------------------------------------------------------------------------------------
1. RX:     0 Kbps      TX:     2 Kbps(    2 Kbps)      CPU:   150K Idle/sec
-------------------------------------------------------------------------------------------
2. CS                :       103 (total)     ¦        34 (1M corr)    ¦        59 (2M corr)      ¦        10 (SAT)
3. SYNC      ( 29%)  :        30 (done)      ¦        73 (miss)       ¦         0 (timeout)
4. SIG       (100%)  :        30 (ok)        ¦         0 (err)        ¦
5. MAC CRC   ( 96%)  :        29 (ok)        ¦         0 (err)        ¦         0 (delimeter error)
6. NDP BA Error      :         0             ¦         0 us (MAX)
7. SNR(1Mhz)         :        27 (pri)       ¦         0 (sec)
8. RSSI              :       -75
9. Avg CFO (Last 16) :      -1.56 ppm
10. CCA              :         0%(1M)        ¦         0%(2M pri)      ¦         0 %(2m sec)
11. Noise (RSSI-SNR) :      -102
12. TX Idle Time     :         0 us          ¦         0 ea
-------------------------------------------------------------------------------------------
```

2.2.21 set deepsleep_gpio <dir> <out> <pullup>

Set the GPIO direction, output data, and pullup settings using a 32-bit mask during deep sleep operation.

Parameters

dir : GPIO direction

out : GPIO output value

pull-up : GPIO pull-up option

```
NRC> set deepsleep_gpio 0x00c00000 0x10203000 0x00000000
OK
```

2.2.22 set support_ch_width [0|1]

Set the supported channel width in S1G capability IE.

Parameters

0:1/2M, 1:1/2/4M

```
NRC> set support_ch_width 1
set support_ch_width : 1
OK
```

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2.2.23 set ampdu_mode [mode]

Set the AMPDU(Aggregate MAC Protocol Data Unit) mode

Parameters

mode : auto, manual or auto.

Returns

```
NRC> set ampdu_mode auto
OK
```

2.2.24 set bcn_mcs [vif id] [MCS]

Set the MCS (Modulation and Coding Scheme) level for beacon transmissions.

Parameters

vif_id : interface ID (default 0, vif_id can be 0 or 1 when the concurrent mode is enabled)

MCS level : 10, 0, 1, 2, 3, 4, 5, 6, 7

Returns

```
NRC> set bcn_mcs 0 5
set bcn_mcs: 5
OK


NRC> show bcn_mcs 0
----------------------- current beacon rate -------------------------
VIF#:0, Current beacon mcs = 5

---------------------------------------------------------------------
OK
```

2.2.25 set rc_param [EWMA] [Update interval] [Probe interval]

set rate control parameters.

Parameters

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EWMA value: 1 means 10%, 2 means 30%, ..., 5 means 50%

Update interval: 1 means 100ms, 2 means 200ms, ..., 7 means 700ms

Probe interval: 1 means 10ms, 2 means 20ms, ..., 255 means 2550ms

Default value : 30% and 500ms and 50ms

Returns

```
NRC> set rc_param 3 6 5
OK


NRC> show rc_param
EWMA(%)                         : 30
Update interval(ms)             : 600
Probe interval(ms)              : 50

OK
```

2.2.26 set cts <on|off>

set CTS on/off (adaptive cts:on, legacy cts:off)

Parameters

on : adaptive cts

off : legacy cts

Returns

```
NRC> set cts on
OK
```

2.2.27 set bgscan_trx [1|0] [wait time operation ch for rx]

set bgscan_trx

Parameters

1|0 : 1:enable or 0:disable

off : wait time operation ch for rx: (0~100)msec

Returns

```
NRC> set bgscan_trx 1 50
set bgscan_trx: 1
OK
```

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2.2.28 set scan_period [dwell time]

set scan period

Parameters

dwell time : dwell time (min 20ms)

Returns

```
NRC> set scan_period 100
set scan_period: 100
OK
```

2.2.29 set mesh_rssi_threshold [threshold value]

set mesh threshold value

Parameters

Threshold_value : mesh_rssi_threshold {-120~-10dBm}

Returns

```
NRC> set mesh_rssi_threshold -40
OK
```

2.2.30 set bcmc_mcs [on|off] [MCS index]

Enables or disables the MCS setting for BCMC(Broadcast/Multicast) frame

Parameters

on : Enable BCMC MCS with the specified MCS index (10|0|1|2|3|4|5|6|7)

off : Disable BCMC MCS setting

Returns

```
NRC> set bcmc_mcs on 7
set bcmc_mcs: 7
OK
```

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2.2.31 set dhcp_mcs [on|off] [MCS index]

Enables or disables the MCS setting for DHCP frame

Parameters

on : Enable DHCP MCS with the specified MCS index (10|0|1|2|3|4|5|6|7)

off : Disable DHCP MCS setting

Returns

```
NRC> set dhcp_mcs on 7
set dhcp_mcs: 7
OK
```

2.2.32 set mgmt_mcs [on|off] [MCS index]

Enables or disables the MCS setting for management frame

Parameters

on : Enable management frame MCS with the specified MCS index (10|0|1|2|3|4|5|6|7)

off : Disable management frame MCS setting

Returns

```
NRC> set mgmt_mcs on 7
set mgmt_mcs: 7
OK
```

2.2.33 set prim_loc [0|1|2|3]

Set the primary location index for 1 MHz channel sniffing

Parameters

0|1|2|3 : Primary location index

Returns

```
NRC> set prim_loc 2
OK
```

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2.2.34 set eapol_mcs [on|off] [10|0|1|2|3|4|5|6|7]

Enable or disable fixed MCS for EAPOL frames and set the MCS index when enabled.

Parameters

on | off: Enable or disable fixed MCS for EAPOL frames

10 | 0~7: MCS index to be applied when enabled

Returns

```
NRC> show rx_clear
0
OK
```

2.2.35 set arp_mcs [on|off] [10|0|1|2|3|4|5|6|7]

Enable or disable fixed MCS for ARP frames and set the MCS index when enabled.

Parameters

on | off: Enable or disable fixed MCS for EAPOL frames

10 | 0~7: MCS index to be applied when enabled

Returns

```
NRC> set eapol_mcs on 7
set eapol_mcs: 7
OK
```

2.2.36 set mgmt_mcs10_permit [on|off]

Enable or disable the use of MCS10 for management frames.

Parameters

on | off: Allow or disallow MCS10 for management frame transmission

Returns

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```
NRC> set mgmt_mcs10_permit on
OK
```

2.3 Test commands

2.3.1 test mcs <value>

Set MCS(Modulation Coding Scheme) index.

Parameters

value : 0~7, 10 (10 can be used only in 1MHz bandwidth)

```
NRC> test mcs 7
OK
```

2.3.2 test country <Country>

Set country for a test command

Parameters

Country : Country Code. KR, US, EU……

```
NRC> test country US
OK
```

2.3.3 test cont_tx <freq> <bw> <mcs> <txpwr> <wave type> | <stop>

Starts or stops a countinous TX

Parameters

freq : frequency (in MHz)

bw : 1m, 2m, 4m

mcs : 0~7, 10 (10 can be used only in 1MHz bandwidth)

txpwr : TX power

Wave type (0 : legacy continuous Tx, 1 : Duty99)

Test sequence

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(1) Set initial country code

```
sudo iw reg set (US/JP /etc)
```

(2) Load firmware on host

(ex)

```
sudo insmod /home/pi/nrc_pkg/sw/driver/nrc.ko hifspeed=20000000 spi_bus_num=0 spi_cs_num=0 spi_gpio_irq=5 spi_polling_interval=0 fw_name=uni_s1g.bin auto_ba=1 listen_interval=1000
```

(3) Run firmware CLI commands after loading

- Set country code : "test country [US|JP|etc]"

```
NRC> test country US
OK
```

- Start continuous Tx : " test cont_tx [freq in MHz] [1m|2m|4m in bw] [mcs] [tx power] [wave type(0|1)]"

```
NRC> test cont_tx 920.0 2m 7 18 0
OK
```

- Stop continuous Tx : "test cont_tx stop"

```
NRC> test cont_tx stop
OK
```

Fixed Parameters

- Test period : Infinite until command “test cont_tx stop”
- Ack policy : No Ack
- CCA : Ignore
- Rate control : Disable
- Interval between packets : 10msec
- Frame Length : 756 (1m BW) / 1651 (2m BW) / 3439 (4m BW) bytes

2.3.4 test sine_tx <freq> <bw> <txpwr> | <stop>

Starts or stops a continuous sine wave transmission

Parameters

freq : frequency (in MHz)

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bw : 1m, 2m, 4m

txpwr : TX power

Test sequence

(1) Set initial country code

```
sudo iw reg set (US/JP/etc)
```

(2) Load firmware on host

(ex)

```
sudo insmod /home/pi/nrc_pkg/sw/driver/nrc.ko hifspeed=20000000 spi_bus_num=0 spi_cs_num=0 spi_gpio_irq=5 spi_polling_interval=0 fw_name=uni_s1g.bin auto_ba=1 listen_interval=1000
```

(3) Run firmware CLI commands after loading

- Set country code : "test country [US|JP|etc]"

```
NRC> test country US
OK
```

- Start Tx of Sine wave : "test sine_tx [freq in MHz] [1m|2m|4m in bw] [tx power]"

```
NRC> test cont_tx stop
OK
```

- Stop Tx of Sine wave : "test sine_tx stop"

```
NRC> test sine_tx stop
OK
```

2.3.5 test freq_bw {freq(in MHz)} {bw(1m|2m|4m)}

Set the operating frequency and bandwidth for RF test

Parameters

freq (in MHz): Target RF frequency (center frequency)

bw (1m | 2m | 4m): Channel bandwidth (1 MHz, 2 MHz, or 4 MHz)value

```
NRC> test freq_bw 920.0 2m
OK
```

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2.3.6 test single_tx {remaining} | {stop} | {freq(in MHz)} {bw(1m|2m|4m)} {mcs} {txpwr} {packet count}

Transmit packets in single TX test mode with specified RF parameters or control the TX operation.

Parameters

remaining: Display remaining packet count during transmission

stop: Stop current TX operation

freq (in MHz): RF frequency

bw (1m | 2m | 4m): Channel bandwidth

mcs (0~7, 10): Modulation and Coding Scheme index

[Start]

```
NRC> test single_tx 920.0 2m 7 18 400
OK
```

[Stop]

```
NRC> test single_tx stop
OK
```

2.3.7 test length [length]

Set the payload length for RF test

Parameters

length: Packet payload length in bytes

```
NRC> test length 512
OK
```

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2.4 GPIO commands

The GPIO CLI commands listed below can be used to configure GPIO options to read from and write to GPIO pins. Note that GPIO command usage is valid for non-dedicated GPIO pins only. Refer to the NRC7394 Technical Reference Manual for the list of dedicated GPIO pins.

2.4.1 gpio read <pin index>

Read the value (0: low / 1: high) from a GPIO pin.

Parameters

pin index: GPIO pin index

```
NRC> gpio read 18
1
OK
```

2.4.2 gpio write <pin index> <value>

Write a value (0: low / 1: high) to a GPIO pin.

Parameters

pin index: GPIO pin index

value: 0: low / 1: high

```
NRC> gpio write 18 0
0x47fbfff3
OK
```

2.4.3 gpio direction <pin index> [direction]

Get or set the direction (0: input / 1: output) of a GPIO pin.

Parameters

pin index: GPIO pin index

direction: set GPIO direction

[Get]

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```
NRC> gpio direction 18
1
OK
```

[Set]

```
NRC> gpio direction 18 1
0x07ffff30
OK
```

※ (For set command only) direction: 0: input / 1: output

2.4.4 gpio pullup <pin index> [pull-up option]

Get or set the pull-up option (0: disable / 1: enable) for a GPIO pin.

Parameters

pin index: GPIO pin index

pull-up option: set pull-up option

[Get]

```
NRC> gpio pullup 7
1
OK
```

[Set]

```
NRC> gpio pullup 18 0
0xc0000080
OK
```

※ (For set command only) pull-up option: 0: disable / 1: enable

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3 Revision History

Revision No	Date	Comments
Ver 1.0	4/5/2023	Initial version
Ver 1.1	9/19/2023	Remove 'auto' parameter for set gi [Added] test country, test cont_tx, show cal_use, show sysconfig, show rc_pf, show rc_param, set ampdu_mode, set bcn_mcs, show bcn_mcs, show xtal_status
Ver 1.2	11/15/2023	[Added] show rc, show app_version, show optimal_channel, set duty_debug, set color, set probe_resp_vendor_ie, set report <on|off>, set deepsleep_gpio, set support_ch_width, set bgscan_trx, set scan_period, set mesh_rssi_threshold
Ver 1.3	4/3/2024	[Updated] show rc_param, set rc_param
Ver 1.3.1	7/16/2024	[Updated] show config
Ver 1.3.2	8/19/2025	[Added] set bcmc_mcs, set dhcp_mcs, set mgmt_mcs, set prim_loc, show clock_count {xtal|lpo}, test sine_tx [Removed] set probe_resp_vendor_ie, set rc_pf, show rc_pf
Ver 1.3.3	3/24/2026	[Added] set bcmc_mcs, set dhcp_mcs, set mgmt_mcs, set prim_loc, show clock_count {xtal|lpo}, test sine_tx show rx_count, show rx_clear, set eapol_mcs, set arp_mcs, set mgmt_mcs10_permit, test freq_bw, test single_tx, test length [Removed] show bcmc, set bcmc
Ver 1.3.4	9/28/2026	[Added] show chmgr, set chmgr, set tpc [Removed] show optimal_channel, set self_config

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